YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 5 OF 10 · 40 MINUTES

The mystery liquid 神秘液体

An indicator's colour shows if a liquid is acid, neutral or alkali; reference samples place it on the pH scale, but not which substance it is.

- I can use indicator and reference samples to classify the scene liquid and estimate its pH.
- I can notice evidence that does not fit and suggest explanations that could be checked.

indicator 指示剂 · **acid** 酸性 · **alkali** 碱性 · **neutral** 中性 · **pH scale** pH值（酸碱度） · **colour chart** 比色卡 · **reference sample** 参照样本 · **consistent** 一致的

THE LESSON

Do Now (Checkpoint)	10	Papers on desks; exactly 10 minutes on the timer; collect, then show the Today slide
Hook	2	Demo: vinegar turns indicator pink, baking soda solution blue-green; students call colours
Learn	5	Acid, neutral, alkali, pH scale and colour chart; safety slide; goggles on; copy colour key
Do	15	Groups add 10 drops of each liquid to indicator cups; record colour, type, pH; match suspects
Think and Reflect	8	Why does the liquid not fit? Pair talk; sticky note idea; evidence log row 5; exit ticket

Hands-on: Groups test 5 reference liquids and the scene liquid in red cabbage indicator; read pH; match to suspects.

Collect or assess: Checkpoint (12 marks) into tracker column Q; results table; evidence log row 5 and tracker; exit ticket; sticky notes.

SAFETY

- Eye splashes: goggles on before opening any liquid; rinse eyes with plenty of water.
- Never taste or drink any liquid. Hot water and the kettle: teacher only.
- Wipe spills straight away; check allergy list (lemon juice); wash hands after.

WATCH OUT

- Test the scene liquid first: it must turn blue-green, not purple. Bottled water for indicator and water.
- Keep checkpoint to exactly 10 minutes. Do not explain why the liquid does not fit or confirm a suspect.

Language: Chinese welcome in talk and answers. Band 1 beside language buddy. Traps: spelling 'alkali'; 'neutral' not 'natural'.

Next week: Week 6 (footprints, incoming teacher): keep Parking Board up; print scene footprint at 100% (24 cm).

PREP FOR ONE CLASS

before the lesson about 84 min (30 could be done by someone else); after about 75 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Buy red cabbage (1/4 head), lemon juice, bar soap, bottled water; check kit 10 min · teacher

THE WEEK BEFORE

- [ ] Check the allergy list for lemon juice; plan for any student affected 2 min · teacher
- [ ] Print: checkpoint x25, worksheet x24, teacher copies, colour pH charts x7 10 min · reprographics
- [ ] Laminate and trim the 6 A5 pH colour charts 10 min · technician

THE DAY BEFORE

- [ ] Label 72 cups: 36 indicator, 30 reference, 6 Evidence 4 (optional) optional 15 min · technician
- [ ] Make red cabbage indicator (hot water, teacher only); strain, label, fridge 25 min · teacher
- [ ] Put up Parking Board header, A3 pH chart and Week 4 powder key poster 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Make soap, baking soda, scene liquids; fill, label, test cups; papers out 15 min · teacher
- [ ] Set out 6 trays, Hook demo, sticky notes, case files and cards 5 min · technician
- [ ] Set incoming teacher's seat: lesson plan and a checkpoint paper 2 min · teacher

AFTER THE LESSON

- [ ] Clean-up: liquids down sink, goggles away, wipe tables, wash hands 3 min · students
- [ ] Rinse cups and droppers or recycle; leftover indicator in fridge (3 days) 10 min · technician
- [ ] Photograph Parking Board; read exit tickets; book Week 6 walkthrough 17 min · teacher
- [ ] Mark checkpoint out of 12, enter tracker column Q, then write handover note 45 min · teacher

PRINT

- Worksheet x24 (2 sheets each, pages 1 and 2 double-sided)
- Checkpoint paper x25 (1 sheet, double-sided; 1 for incoming teacher)
- pH colour chart x6 (A5 landscape, colour, laminated)
- pH colour chart x1 (A3 landscape, colour if possible, for the wall)
- Ideas Parking Board header x1 (A4 landscape)
- Lesson plan (one page) x1 for the incoming teacher's seat
- Teacher copies: kit card (A5, 2 sides), mark scheme, handover form

GET OR MAKE (NEW THIS LESSON)

- Small clear plastic cups x74 (72 on trays, 2 for demo)
- Droppers x36 (6 per group)
- Safety goggles x24, trays x6
- Sticky labels x73, sticky notes x24
- Plain white A4 paper x6, A5 laminating pouches x6
- Kettle, heatproof jug, sieve, clean bottle (1 each)
- Kitchen tools: knife, board, grater, spoons, jugs
- Red cabbage x1/4 head
- Lemon juice 60 ml; vinegar 60 ml plus a little for demo
- Plain bar soap x1 (not washing-up liquid), baking soda
- Bottled or purified water 560 ml
- Cloths and paper towels (number not stated)
- Timer, camera or phone, fridge space, board wall space
- Make: indicator 500 ml, soap solution 200 ml
- Make: baking soda solution; scene liquid about 60 ml
- Rinse water cups x6, if 2 droppers per group (optional)

Have ready from earlier weeks: Class box, Case files x24, Scientist's Logbooks x24, My Science Dictionary x28 (4 spare), Role cards incl. language buddy x6 sets, Sentence frame card sets x6, poster, Week 4 powder key poster (A3), Pencils x24

Chinese: draft, needs native-speaker check